e) Each elementary matrix is invertible.

$\boxed{\text{Solution:}}$ On Page 109, from the topmost grey box we know that the given statement is True.

10) a) A product of invertible $n \times n$ matrices is invertible, and the inverse of the product is the product of their inverses in the same order.

$\boxed{\text{Solution:}}$ By Theorem 6(b) on Page 107, it is mentioned that if A and B are $n \times n$ invertible matrices, then so is AB, and the inverse of AB is the product of the inverses of A and B in the reverse order. That is,

1 $\boxed{(AB)^{-1} = B^{-1}A^{-1}}$. By the List of Warnings on Page 100, $(A^{-1}B^{-1}) \neq B^{-1}A^{-1}$. $\therefore (AB)^{-1} \neq A^{-1}B^{-1}$. The given statement is False.

2 b) If A is invertible, then the inverse of A^{-1} is A itself.

$\boxed{\text{Solution:}}$ By Theorem 6(a), the given statement is True.

2 c) If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad = bc$, then A is not invertible.

$\boxed{\text{Solution:}}$ By Theorem 4 on Page 105, if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad - bc = 0$, i.e. $ad = bc$, then A is not invertible. $\therefore$ The given statement is True.

d) If A can be row reduced to the identity matrix, then A must be invertible.

$\boxed{\text{Solution:}}$ By Theorem 7 on Page 109, the given statement is True.

e) If A is invertible, then elementary row operations that reduce A to the identity I_n also reduce A^{-1} to I_n.

$\boxed{\text{Solution:}}$ By Theorem 7 on Page 109, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1}. We shall get I_n from A^{-1} by the reverse of the row operations that transformed I_n into A^{-1}. $\therefore$ The given statement is False.

11) Let A be an invertible $n \times n$ matrix, and let B be an $n \times p$ matrix. Show that the equation $AX = B$ has a unique solution $A^{-1}B$.